Advances in Algorithms (CS503)

Mid Semester Examination, September 2018

Department of CSE, IIT Patna

Total Marks: 60 Duration: 120 min

Please give concise answers to all the questions. Unnecessary lengthy answers will attract deduction of marks.

1. Answer the following questions: [6+6+3]

   I. Give three non-optimal greedy strategies for interval scheduling problem and provide counter-examples for their non-optimality.
   II. Prove the correctness of the optimal greedy algorithm for Interval scheduling problem.
   III. Prove that a forest has exactly (***n-m***) number of trees where ***n*** is the number of nodes and ***m*** is the number of edges in that forest.

2. Answer the following questions: [4 +5+6]

   I. What is pseudo-polynomial algorithm? Give an example and show why it is pseudo-. polynomial.
   II. What are the differences between Top-down DP (Memoization) and Bottom-up DP (Tabulation)? Give examples of the both.
   III. Give a branching strategy and a bounding strategy for solving 0/1 knapsack problem using Branch and Bound approach.

3. Answer the following questions: [7 +6+2]

   I. A full "dynamic table" creates a new table of twice size and copy all the existing elements in the new table when an add_element operation is called. Now consider a dynamic table initialized with size 1 and a sequence of *m* add_element operations on it. Analyze the amortized cost per operation. Here *m* is unrestricted. Use Potential method and Aggregate method.
   II. Describe Max-Flow Min-Cut theorem. Write Ford-Fulkerson algorithm including augmenting method?
   III. What is ploy-time reduction? Why do we use it?

4. Answer the following questions: [6+6+3]

   I. Prove that finding a clique of size ***k*** in a graph ***G*** is a NP-complete problem. Use poly time reduction from 3CNF-SAT problem.
   II. Prove that greedy algorithm (list scheduling) for Load Balancing problem is a 2-approximation algorithm. You have to prove the required lemmas as well.
   III. How to solve subset sum problem approximately? Write in brief.